Dual-ligand fluorination

01 / Experiment

Use the two-page bench sheet at the robot. Keep the protocol and its generated CSVs together; `dosing.csv` gives well identities and `deck_loads.csv` gives stage-specific source fills.

Reaction and design

Pyridine-2-sulfonyl fluoride (PyFluor) to 2-fluoropyridine. Per vial: 25 µmol SM, 1.5 µmol each ligand, 3 µmol Pd(COD)(DQ), 2.5 µmol mesitylene; 250 µL toluene, 95 °C, 500 rpm, 4 h under air. No base or external fluoride.

Round one: **48 distinct pairs, all 30 available ligands**; L07 is unavailable. Round two: **22 new pairs + L17 tBuBrettPhos at 12 mol% total ligand**. No repeats. A better pair yield establishes improvement against this reference; it does not establish reproducibility or ligand synergy.

Before the clock starts

- Fit **p300 single GEN2 left; p300 eight-channel GEN2 right**. Reaction vials have 13 mm pitch: use the single only. The multi serves the standard 96-well LC preparation plate.
- Check the supplied block on the flat adapter: secure mounting, latch closure, cap clearance within **45 mm**, offsets and thermal transfer. Pipette **1.5 mm above the bottom** by default; check actual immersion. Caps are manual; no stir bars or gripper.
- Confirm the existing **1.7-min full LC cycle**, method ID, injection volume, UV channel/wavelength and authentic-standard RT/LOQ settings. Requested 210 nm is provisional; the old example wavelengths are not our method. Record local evidence before release.
- Prepare stocks, caps, tips, 71 fresh reaction vials, two final autosampler-compatible 96-well deep-well LC plates with pierceable seals, plus standards. The supplied VWR definition is 2.2 mL/well; confirm the physical plate fits BOTH the flat adapter and autosampler, including seal, carrier and needle depth. The derived `_hs.json` adds movement geometry only; it does not establish physical compatibility. App-simulate the released Python. Open setup details.

Slot	Load
1 / 4	2 mL Eppendorf working tubes; ligands at CSV addresses, L07 absent; combined SM/IS in **slot 4 D2**
2	A2 Pd; quench A4/A6/A9; diluent A5/A7/A10/A11/A12. A1/A3/A8 unused
3 / 10	Reaction block: staff move it by hand at App prompts; pipetting / heater-shaker
6	VWR 96-well 2.2 mL deep-well preparation plate, or matching configured replacement; fill here, shake on 10, then directly to HPLC
5 / 8 / 9	Single tips / multi tips / multi tips. Keep slots 7 and 11 empty

Dose and react

1. Robot doses **25 µL combined 1 M SM +100 mM mesitylene in toluene** per open vial. The 25 µL remains in the reaction. Load **1.31 mL** in the **2 mL source tube, slot 4 D2**, and preserve the off-deck R2 reserve.
2. Add **75 µL ligand A +75 µL ligand B +75 µL Pd =250 µL total including the SM/IS stock**. Ligands are **20 mM**, Pd **40 mM**. Reference: **150 µL L17** instead of the two ligand portions. No separate IS or solvent top-up.
3. Hand-tighten all caps. Move/latch at local pauses; run **95 °C, 500 rpm, 4 h**. Every block move, cap change and tip-rack replacement is manual.
4. During hold one, run **14 calibration injections**. Hand-dose **23 labelled fresh vials with 25 µL combined SM/IS stock each** from the off-deck reserve; **cap promptly and retain the toluene**. Add catalysts only after R2 selection. Reuse the same block.

12-hour operating target: 480 min reaction holds +120.7 min LC; **119.3 min for all remaining handling, workups, decisions, final analysis and contingency**. Calibration and R2 SM predosing overlap hold one. After R1 sampling, prepare the R2 block while LC runs; catalyst dosing waits for all 48 results and selection. One LC plate per round, four moves total; no R1 split. Confirm the full cycle and handling times locally before release.

Workup and LC sampling

1. **Cool sealed vials off the heater before opening.** The App separately waits for module temperature <=30 °C before workup; its reading does not establish vial temperature. Open for addition; add **500 µL 9:1 ACN/water +0.15% formic acid**. Nominal extract: 750 µL, 0.1% acid, 1.25 mL headspace. Quench and diluent contain no IS.
2. **Hand-recap every vial.** Shake **1500 rpm, 2 min**, heater off; return to slot 3. The block must have cooled for workup.
3. **One LC plate per round.** Let bubbles settle after the automated extract shake; open the indicated reaction vials for sampling. Other vials stay capped.
4. **Robot dilution:** fresh tip per sample; two slow same-well prewet cycles, without blowing air into the extract. Aspirate 21 µL at 1.5 mm, retain 1 µL reverse surplus, add 2 µL air gap and dispense nominal **20 µL into 980 µL LC diluent**, noncontact. Discard the tip; never revisit the source after destination contact.
5. **Recap sampled reaction vials.** The reaction block stays in slot 3. At the App move prompt, take the completed LC plate **slot 6 → cooled heater-shaker slot 10**. Keep A1 orientation. Robot latches and shakes the plate **UNCOVERED, heater OFF, 1000 rpm for 2 min**; use the locally confirmed splash-free speed. It waits for module temperature ≤30 °C before loading. No manual sample mixing/pipetting.
6. **Move directly 10 → HPLC** at the prompt; apply the confirmed pierceable autosampler seal. Load the correct carrier, plate definition and needle depth. R1 plate **R1-LC01: A1-H6, 48 samples**; R2 **R2-LC01: A1-H2 plus A3-G3, 23 samples**, column-wise. **H3 contains diluent only: do not inject.** Use `lc_plates.csv` and `lc_sequence.csv`; the 48-vial reaction map differs from the LC map. Inject directly from the plate; no individual sample-vial transfers.

Final sample: 1.00 mL; maximum SM/product **666.7 µM**; mesitylene **66.7 µM**; formic acid **0.1%**; toluene **0.667%**. Set injection volume from the actual calibrated method. Final LC plates and their headspace remain unchanged during injection; seal after shaking to limit evaporation.

Pipetting and transfer faults

Condition fresh dosing tips twice in their own stock. Use slow aspiration, short settling delays, a small air gap and reverse surplus for the organic stocks. Keep active sources covered between stages and bulk reserves capped. Follow CSV fills and dead volumes; never return working liquid to bulk stock.

Reuse a tip only for consecutive **same-stock, noncontact** dosing above the destination top. Different ligands and all analytical samples use fresh tips. Avoid deep immersion, walls and bottoms. Repeated aspiration does not dissolve an insoluble stock and may introduce bubbles.

Pause for persistent layers, solids at the aspiration point, bubbles, air uptake, dripping or tip contact. Do not sample an arbitrary layer. Do not filter, centrifuge or decant without recording its effect on analyte/IS recovery. High rpm alone does not establish quench completeness or extraction recovery; no separate homogeneity study is planned.

For an uncertain LC dilution, remix the original extract and prepare a fresh documented dilution. Do not guess a top-up. Retain rejected measurements with a reason and accept one result per sample; do not select the highest yield.

Changeover and finish

During R1 LC, after all R1 sampling is complete, archive those vials and fit the 23 capped liquid-predosed vials into the same block; verify orientation. After all **48 R1 results** pass QC, the pipeline selects **22 new pairs +L17 reference**. The R2 pause confirms this changeover; do not replace already-loaded new vials again. Refill R2 deck loads and **remove all 23 caps before dosing**. **No SM or IS redosing. RockPhos L03 is limited to four R2 wells.** Repeat reaction/workup.

Stop the clock after all 23 final injections and quantitative analysis. Archive source exports, maps, assay/stock records, method/channel, calibration, QC, times, scoring/cache records, final CSVs, matching protocols and robot JSON logs. Keep experimental data private pending team agreement.

02 / Stocks and calibration

Use `stock_preparation.csv` for masses/reserves; load only the stage's `deck_loads.csv` quantities. **Bulk volume is not deck fill.** Correct masses for COA assay. Prepare ligands in the glovebox.

Reaction stocks

Stock	Bulk preparation at 100% assay	Per reaction
SM + IS	**2.0 mL, 1 M SM +100 mM mesitylene in toluene:** 322.32 mg SM +24.04 mg mesitylene; make to FINAL volume in a capped 4 mL glass vial	**25 µL**, retained; supplies both SM and IS
Each ligand	**20 mM in toluene**; masses/reserves and 4 or 8 mL bulk-vial sizes in CSV	**75 µL each**; reference 150 µL L17
Pd(COD)(DQ)	**40 mM in toluene**; current plan 10 mL, 151.52 mg	75 µL
RockPhos, L03	**800 µL, 20 mM:** 7.50 mg at 100% assay; 7.73 mg at 97%. 4 mL bulk vial; **10 mg available, at most 4 R2 wells**	75 µL; two frozen R1 wells

Use capped glass bulk vials and **2 mL Eppendorf working tubes, max 1.95 mL**, in plastic racks. Combined stock: **slot 4 D2, load 1.31 mL for R1**; keep **575 µL off deck for R2**. The 2 mL preparation rounds the requested ~10% allowance; includes dead volume/surplus. Ligands retain 50% margin. Refill from reserves; never return working liquid to bulk.

Workup solvents - no internal standard

Solution	Prepare	Use
Quench	**100 mL, 9:1 ACN/water +0.15% v/v formic acid:** 150 µL acid, make to volume with premixed solvent	500 µL; recap, 1500 rpm, 2 min
LC diluent	**125 mL, 1:1 water/ACN +0.1% v/v formic acid:** 125 µL acid, make to volume with premixed solvent	980 µL +20 µL extract

Analytical stocks - matched matrix

Matrix, 100 mL: 0.6667 mL toluene +50.1492 mL ACN +0.1000 mL formic acid; water to volume. Approximate diluted-extract matrix; ignores volume contraction. Prepare all analytical stocks in it.

Use `analytical_stock_preparation.csv`. Prepare **two independently weighed 2 mL, 10 mM masters per analyte**: **SM 3.22 mg each; product 1.94 mg each**, corrected for assay. Use 4 mL glass vials, no IS. One master fits calibration; the other supplies the check/optional pilots. Save unused stock; increase minimum weighing if required by the balance.

1 mM working stocks: 20 µL FIT master +180 µL matrix; 0.5 mL tube. Use the LC instructions. Prepare standards manually; use authentic SM/product masters separately from the combined reaction stock.

Analytical mesitylene, 5 mM, 20.00 mL: 12.019 mg in matched matrix, separate from the combined reaction stock.

Calibration during hold one - 14 injections

Prepare seven **1 mL standards per analyte** in a compatible plate/vials. Each receives **13.333 µL of 5 mM mesitylene**. Add analyte/matrix below; mix, seal and use `calibration_template.csv` IDs.

Analyte, µM	Role	Stock, mM	Analyte, µL	Matrix, µL
0	Fit blank	None	0	986.667
10	Fit	1	10	976.667
50	Fit	1	50	936.667
150	Fit	10	15	971.667
350	Fit	10	35	951.667
800	Fit	10	80	906.667
666.667	Independent check	10, independent	66.667	920.000

Fit blank +five levels; exclude the check. Require **R^2 >=0.995**, **check error <=15%**, measured separation/LOQs. Record method, wavelength and injection volume; lowest standard is not automatically the LOQ.

03 / Local pipeline

The **initial 48 pairs are ready** in `campaigns/hackathon2026/round1_design.csv`. Local Python converts integrated UV areas to calibrated yields; Claude selects the next conditions. Staff use the bench sheet.

Set up before the clock

Use Python 3.12: `uv sync --python 3.12 --extra sources`, then prefix commands with `uv run`. Alternatively activate a venv and install `python -m pip install -e '.[sources]'`.

```
mkdir -p local_data
cp -n hte_inputs/campaign.json local_data/campaign.json
cp -n hte_inputs/calibration_template.csv local_data/calibration-plan.csv
cp -n anthropic.key.example anthropic.key
chmod 600 anthropic.key
```

Use **schema 3**; replace older preparation settings/protocols, preserving measured LC facts.

Put the raw Claude key in `anthropic.key`; preserve any existing key. Never send it to the robot, Git or email.

In `local_data/campaign.json`, set actual method/channel, full LC cycle, measured reference RTs/LOQs and applicable release evidence. Confirm final-plate shaker/autosampler compatibility and uncovered shake speed; confirm the measured handling plan within the 119.3-min remaining budget before release. Enter measured stage totals under `timing` (both rounds combined); `check` separates the target from those allowances. HTEL setup facts.

```
python -m hte.cli --config local_data/campaign.json check
python -m hte.cli --config local_data/campaign.json export \
  --design campaigns/hackathon2026/round1_design.csv \
  --output output/r1-dose --live --stage dose_and_react
```

App-simulate and start locally with matching CSVs. Omit `--live` for guarded previews. R1 dosing may precede calibration once other checks pass.

During hold one

Run/import **14 calibration injections** using the LC instructions. Staff predose 23 fresh vials with 25 µL combined SM/IS in toluene and cap promptly. After calibration passes, set `lc_calibration_validated=true`; export workup:

```
python -m hte.cli --config local_data/campaign.json export \
  --design campaigns/hackathon2026/round1_design.csv \
  --output output/r1-workup --live --stage workup
```

After all 48 R1 results

Default: **one final LC plate/export per round, four moves total; no R1 split**. Robot dilutes and shakes. Use `lc_plates.csv` and complete `lc_batchNN_injections.csv` with actual method/sequence identities. Each export needs its own map; match glob order and exclude unrelated files.

```
python scripts/hte-lc-results.py next \
  --dosing output/r1-dose/dosing.csv \
  --exports local_data/r1-batch*.csv \
  --injections local_data/r1-map*.csv \
  --output output/r1-decision
```

Calibration defaults to `local_data/calibration.csv` (`--calibration` overrides). After all 48 results pass QC, Claude scores **387 untested pairs** and selects **22 new pairs +L17 reference**. Both prompts include 17 SI yields. R2 uses no embedding selector; L03 is capped at four wells.

The printed decision folder holds `next_design.csv`, `next_dosing.csv`, analysis, scores and `protocols/`. Use released Python and R2 deck loads. If checks remain open, export `next_design.csv` with `--history output/r1-dose/dosing.csv`, a fresh folder and `--live`. Fit the 23 capped liquid-predosed vials; **no SM/IS redosing**. RockPhos L03 may occupy at most four R2 wells.

After all 23 R2 injections, run the helper with `final`, R2 `dosing.csv`, the matching plate export/map and a new output folder. This calculates final results without Claude. Compare pair yields with the measured L17 reference and retain run times.

Practice and recovery

Practice: `python -m hte.cli practice-analytics --output output/pilot`; real pilot replay. `rehearse-loop` also mocks feedback; no paid calls. Use fresh import folders and review interrupted scoring before retrying.

195 tests passed; protocols simulated. Validation, ML strategy, PDF updates.